How to Help Transformers Count

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Introduction

Problem: Transformers see text, not structure.

Gap: Without explicit understanding of structure, transformers struggle with tasks that require structured reasoning, like arithmetic.

Hypothesis: Transformers will learn arithmetic better with number representations that explicitly encode digit position (Figure 1).

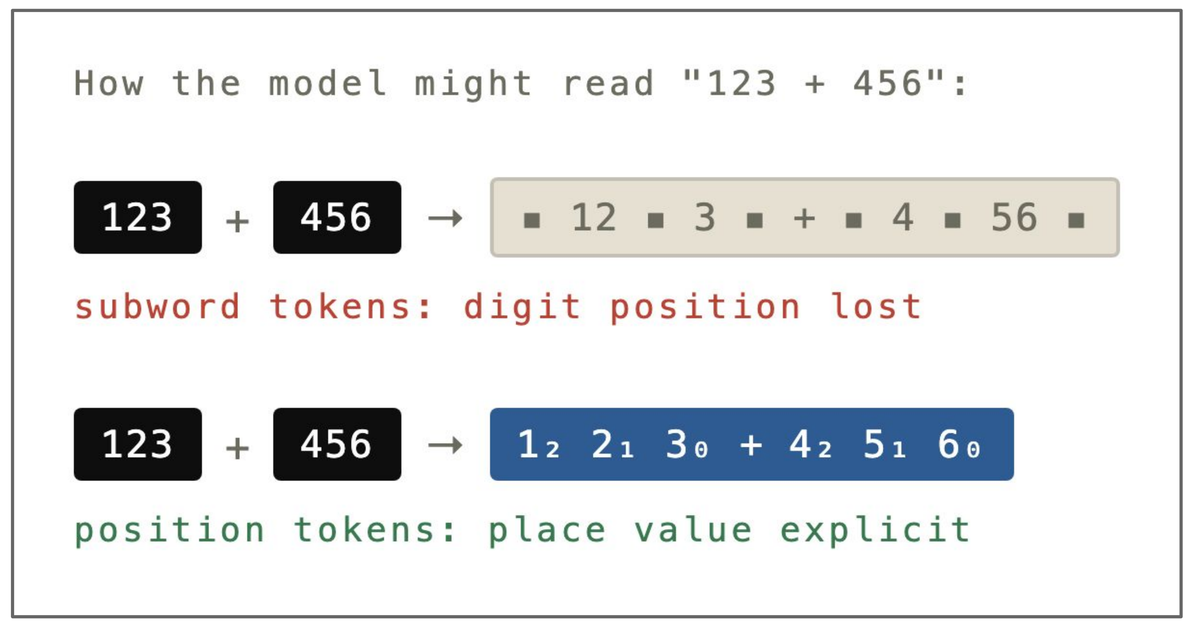

Figure 1. Subword Token v.s. Position Token.

We attempted to reproduce the findings from "*Investigating the Limitations of Transformers with Simple Arithmetic*" (Nogueira et al., 2021), summarized by Figure 2.

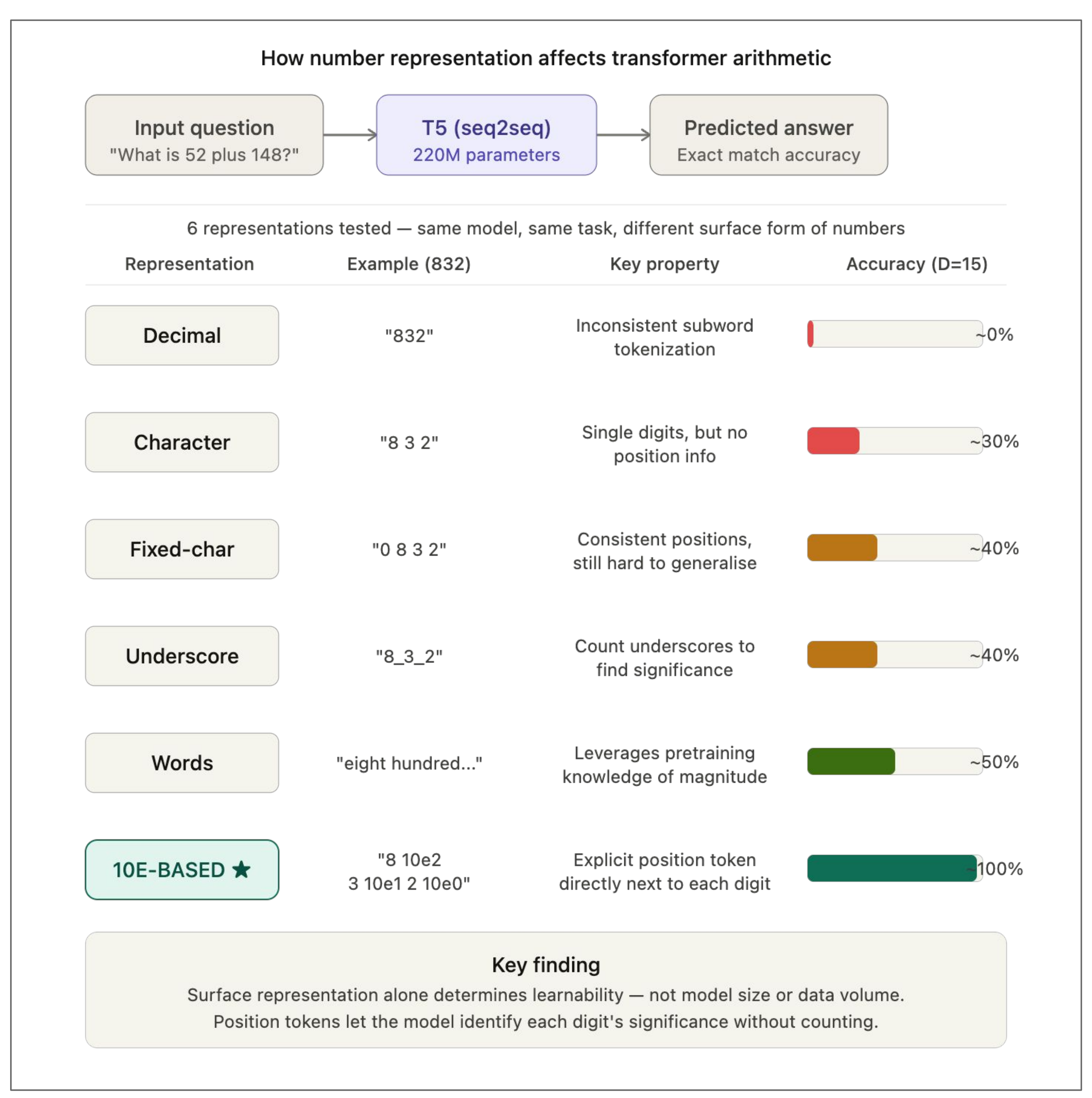

Figure 2. Summary of Nogueira et al., 2021 Findings.

Methodology

Models: T5 (Raffel et al., 2020), specifically T5-small (T5-60M) and T5-base (T5-220M).

As shown in Figure 3, for each model:

- Train on one of the 7 representations (Figure 5) on a value of D ∈ {2, 5, 10, 15, 20, 25, 30}.
- Test & compute accuracy (1 if output exactly matches target output, 0 otherwise).

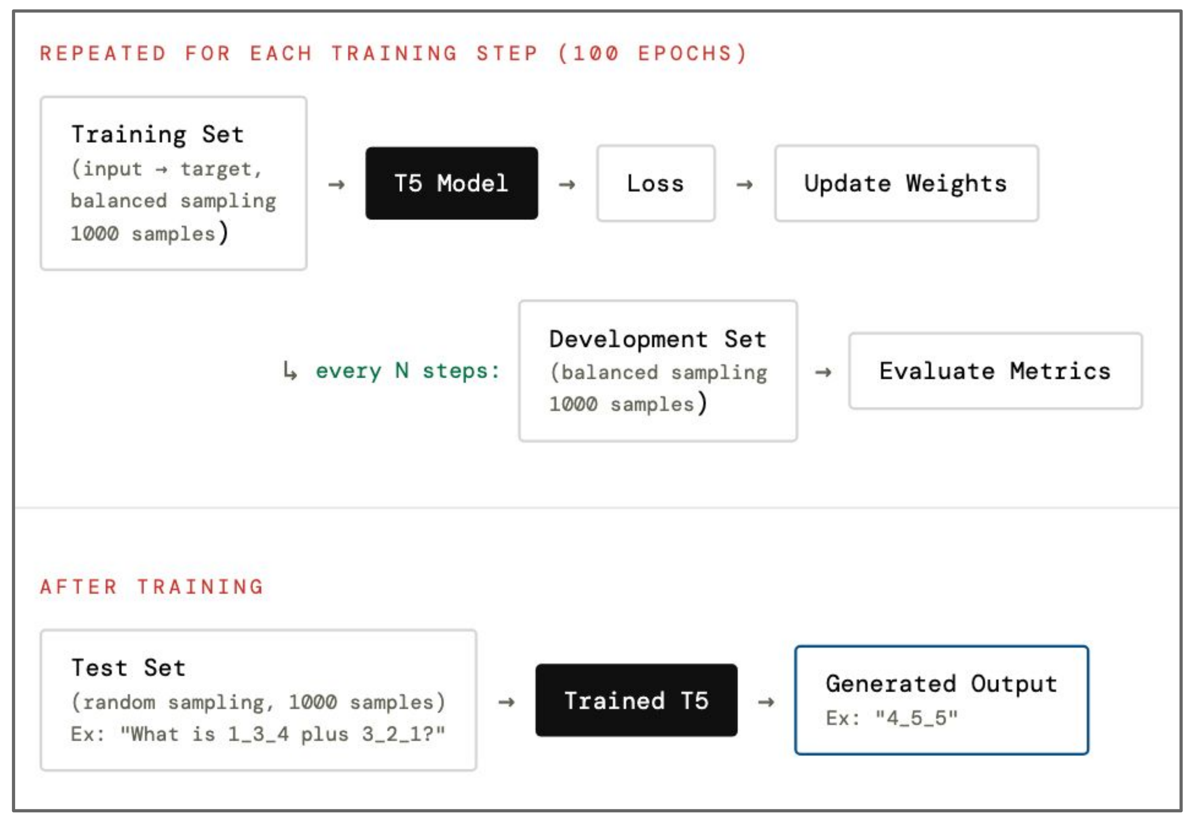

Figure 3. Flow diagram of model training & testing.

The training and development sets were generated using balanced sampling, while the testing set was generated using random sampling (Figure 4).

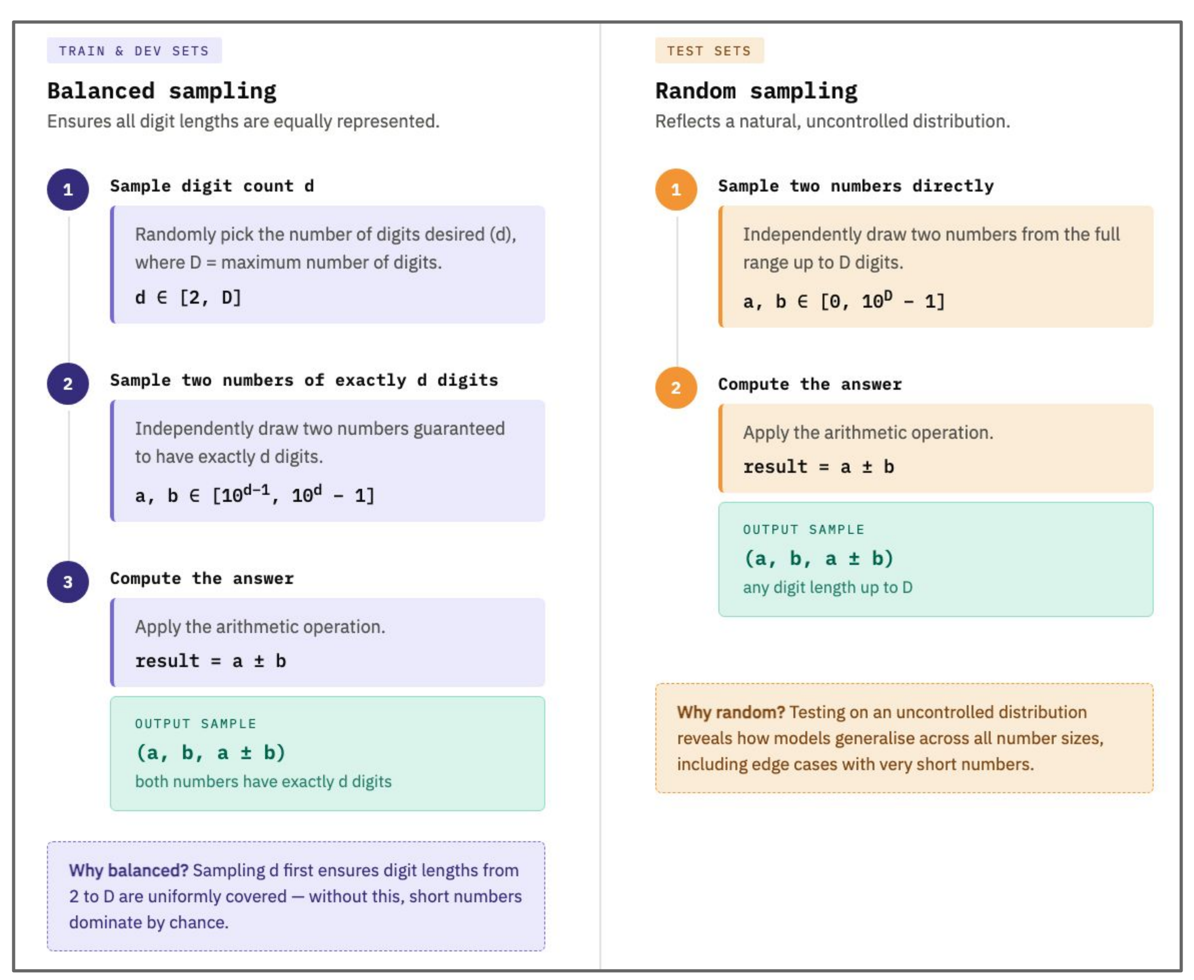

Figure 4. Balanced Sampling v.s. Random Sampling.

WITH POSITION TOKENS	
10ebased	1 10e2 2 10e1 3 10e0 + 4 10e2 5 10e1 6 10e0
10based	1 100 2 10 3 1 + 4 100 5 10 6 1
SURFACE FORM ONLY	
words	one hundred twenty three + four hundred fifty six
underscore	1_2_3 + 4_5_6
fixed char	1 2 3 + 4 5 6
character	1 2 3 + 4 5 6
decimal	123 + 456

Figure 5. The 7 Number Representations.

Additional Experiments: We compared model performance on addition versus subtraction tasks under identical training settings. We hypothesized that subtraction will be more difficult as the model is required to learn sign handling.

Results

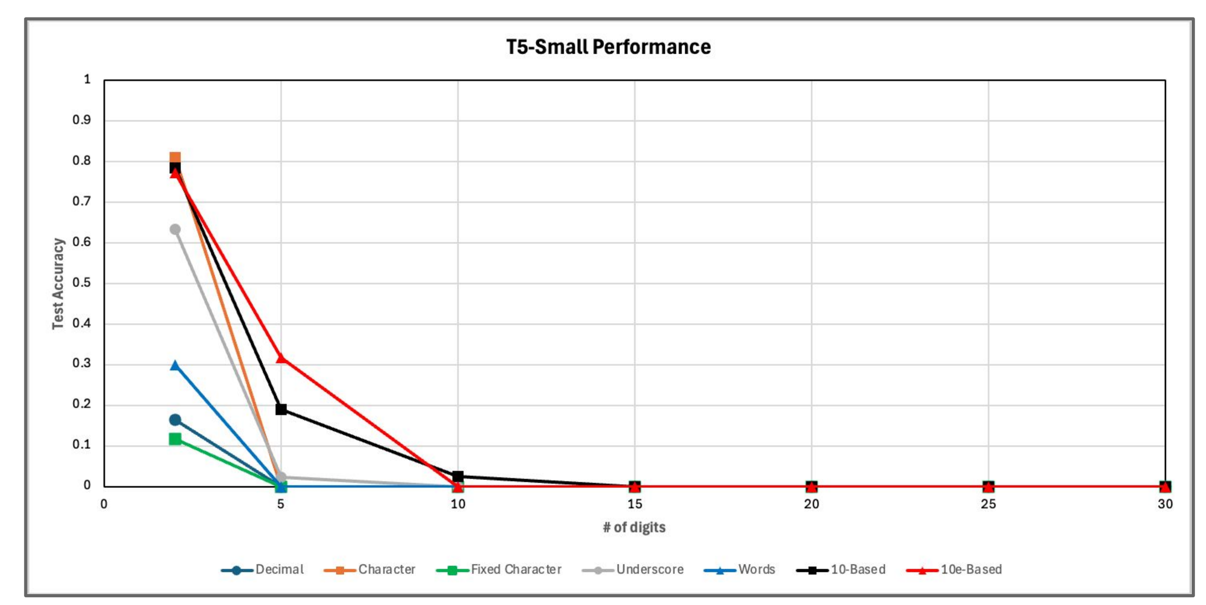

Figure 6. Performance of Representations on T5-Small.

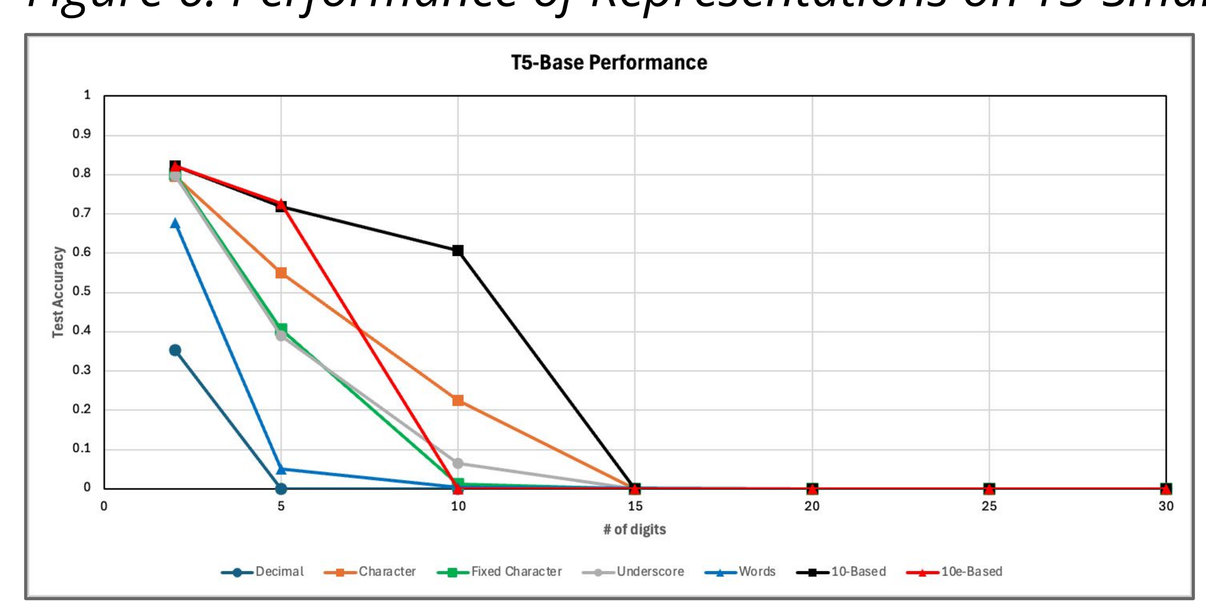

Figure 7. Performance of Representations on T5-Base.

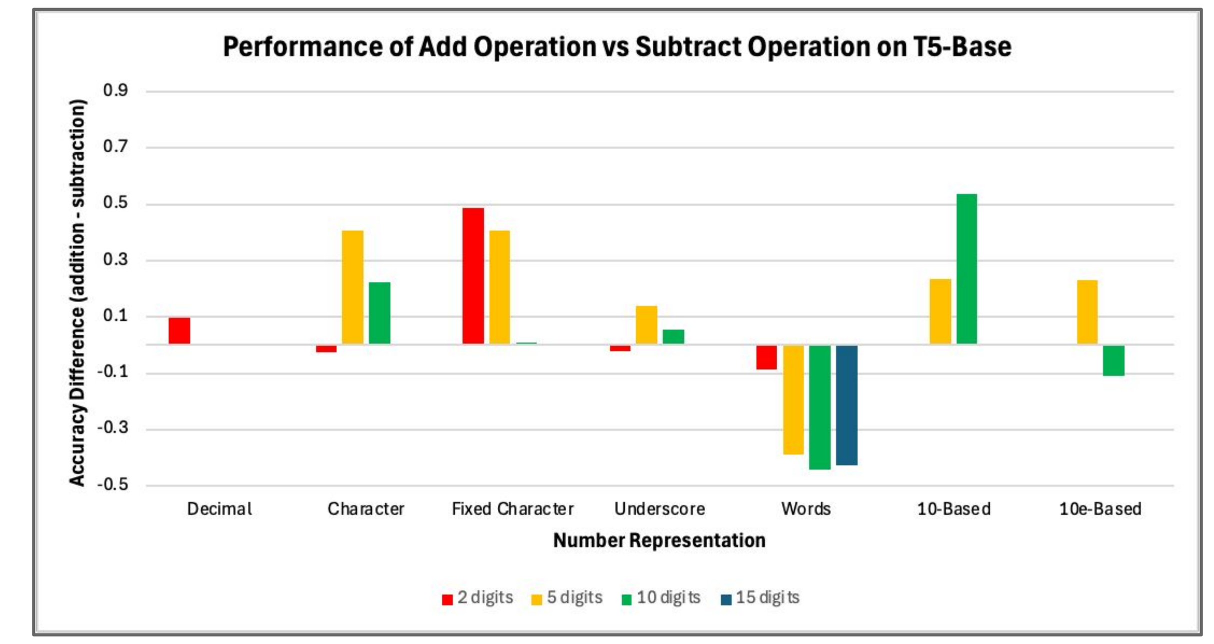

Figure 8. Addition Task v.s. Subtraction Task Accuracy.

Conclusion

Trend Match: Our small-scale replication successfully reproduced the paper's qualitative results (Figures 6, 7).

Tokenization: Position tokens outperformed subword methods by allowing the model to identify each digit's significance directly.

Scaling: Google Colab Hardware memory limits constrained T5-Base performance beyond 15 digits, compared to the paper's model performance.

Arithmetic: Subtraction is harder than addition (borrowing, order sensitivity), but word-based representations may help through explicit sign encoding (Figure 8).

Future Work

We can extend our experiment as follows:

- Investigate multiplication and division.
- Test extrapolation, training and testing on different values of D to see if the model can generalize beyond its training distribution.
- Explore whether a mix of representations during training helps generalization.

References

1. Nogueira, R., Jiang, Z., & Lin, J. (2021). *Investigating the limitations of transformers with simple arithmetic tasks*. arXiv. https://arxiv.org/abs/2102.13019
2. Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., & Liu, P. J. (2019). *Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer*. arXiv:1910.10683. https://arxiv.org/abs/1910.10683

All diagrams in the Introduction and Methodology sections were generated with Claude.